

EE6406

Analytic and Ensemble Machine Learning

(Lecture 6: Advanced Analytic Classification)

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EE6406: Analytic and Ensemble Machine Learning

Lecture 6

Advanced Analytic Classification

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Introduction

- Apart from the error magnitude based methods covered in previous lecture, a more matching approach for classification is to **directly formulate the classification metric as the learning objective**.
- This approach has been a classical subject of study known as *classification error (CE) based methods*. However, due to the **counting** nature of the classification metric (e.g., by accumulating the number of successful prediction counts), attempts to the problem had inevitably resulted in solving a **nonlinear formulation**.
- In this lecture, we introduce two **classifier learning solutions** which can be obtained **in closed form** and yet possesses nonlinear mapping capability. The first method directly uses the **total error rate (TER)** as the learning objective function while the second method uses **area under the ROC curve (AUC)** as the learning objective function.

Total Error Rate Based Learning

The **total error rate (TER = FPR + FNR)** would be a suitable candidate for objective function formulation since it is directly accumulating the **classification error count in normalized form**. Let's denote the positive and negative examples of variables for $\mathbf{x}$ and m by the superscripts $+$ and $-$, respectively, and by expressing in terms of classification based on a threshold, τ , the TER can be written as:

$$\begin{aligned} \text{TER} &= \text{FPR} + \text{FNR} \\ &= \frac{1}{m^-} \sum_{j=1}^{m^-} \text{cls}(g(\mathbf{x}_j^-, \mathbf{w}) \geq \tau) + \frac{1}{m^+} \sum_{i=1}^{m^+} \text{cls}(g(\mathbf{x}_i^+, \mathbf{w}) < \tau), \quad (5.65) \end{aligned}$$

where $\text{cls}(\cdot)$ is the classification function given in (5.68) which is defined for a particular threshold τ . In other words, $\text{cls}(g(\mathbf{x}_j^-, \mathbf{w}) \geq \tau) = 1$ whenever $g(\mathbf{x}_j^-, \mathbf{w}) \geq \tau$ holds true and $\text{cls}(g(\mathbf{x}_j^-, \mathbf{w}) \geq \tau) = 0$ otherwise. Similarly, $\text{cls}(g(\mathbf{x}_i^+, \mathbf{w}) < \tau) = 1$ whenever $g(\mathbf{x}_i^+, \mathbf{w}) < \tau$ holds true and $\text{cls}(g(\mathbf{x}_i^+, \mathbf{w}) < \tau) = 0$ when $g(\mathbf{x}_i^+, \mathbf{w}) < \tau$ is not true.

Total Error Rate Based Learning

By noting the **symmetry between $cls(g(\mathbf{x}_j^-, \mathbf{w}) \geq \tau)$ and $cls(g(\mathbf{x}_i^+, \mathbf{w}) < \tau)$** , a simple change of variables can unify the two classification functions $cls(g^- \geq \tau)$ and $cls(g^+ < \tau)$ into one loss form. Define $\epsilon_j = g(\mathbf{x}_j^-, \mathbf{w}) - \tau$ for $j = 1, 2, \dots, m^-$ and $\epsilon_i = \tau - g(\mathbf{x}_i^+, \mathbf{w}) - \Delta$ for $i = 1, 2, \dots, m^+$. Here, $\Delta \rightarrow 0$ which is to account for the strict inequality in FNR of (5.65) can be ignored in practice. Then the TER loss function can be expressed in terms of ϵ_j and ϵ_i which has a threshold at zero, giving

$$\text{TER} = \frac{1}{m^-} \sum_{j=1}^{m^-} L(\epsilon_j \geq 0) + \frac{1}{m^+} \sum_{i=1}^{m^+} L(\epsilon_i \geq 0). \quad (5.66)$$

Total Error Rate Based Learning

The goal to train a predictor as a classifier is then to find the minimizer of TER given by

$$\begin{aligned}\hat{\mathbf{w}} &= \arg \min_{\mathbf{w}} \text{TER} \\ &= \arg \min_{\mathbf{w}} \left\{ \frac{1}{m^-} \sum_{j=1}^{m^-} L(\epsilon_j \geq 0) + \frac{1}{m^+} \sum_{i=1}^{m^+} L(\epsilon_i \geq 0) \right\}. \quad (5.67)\end{aligned}$$

Here, **loss function** $L(\cdot)$ can be considered an alternative version of the **classification decision function** (5.68) with positive and negative targets ($\{-1, +1\}$ instead of $\{0, 1\}$) **thresholding at $\tau = 0$** .

$$cls(g(\mathbf{x}_j)) = \begin{cases} 1 & \text{if } g(\mathbf{x}_j) \geq \tau \\ 0 & \text{if } g(\mathbf{x}_j) < \tau \end{cases}, \quad j = 1, 2, \dots, n. \quad (5.68)$$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1

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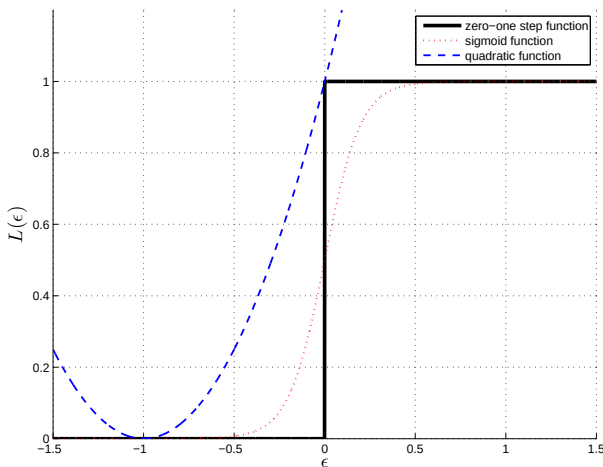


Figure 1: Sigmoidal and quadratic approximations to the zero-one step loss function. The sigmoidal function is adopted as $1/(1 + \exp(-10\epsilon))$

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Total Error Rate Based Learning

- Fortunately, the optimization problem can be approached from [matching the link-loss functional pair](#) perspective.
- Mathematically, the TER formulation in (5.65) or (5.67) constitutes a *functional*¹ since it is a [composite function](#), i.e., $L(g(\cdot))$, where L is a loss function and g is a link function.
- By [matching the linear link](#) ($g(\mathbf{x}, \mathbf{w}) = \mathbf{w}^T \mathbf{x}$) with a [quadratic loss](#) ($L(\cdot) = (\cdot)^2$), a [convex formulation](#) can be obtained. Particularly, in this linear-quadratic pair, a [closed-form solution](#) can be permissible.
- However, we recall that the quadratic function is by no means closed to the loss property of having a heavier penalty for an incorrect output than a correct output where a zero-one function possessed. In other words, a monotonic loss function is necessary for the error counting loss formulation.

¹In mathematics, a *functional* is often referred to as a *function of a function*. ▶

Total Error Rate Based Learning

Nevertheless, from data perspective, we can seek to **manipulate the data such that only one quadratic arm is utilized during estimation.**

To achieve this goal, **a uniform offset** can be introduced to the entire bulk of input data so that they are **projected onto a single quadratic arm.** The dashed curve in Fig. 1 illustrates a shifted quadratic function plotted along with the zero-one step loss function. Translating this idea to our error rate loss function, the following quadratic approximation to (5.67) can be adopted in our minimization formulation:

$$\min_{\mathbf{w}} \text{TER}(\mathbf{w}) \approx \min_{\mathbf{w}} \left\{ \frac{1}{2m^-} \sum_{j=1}^{m^-} (\epsilon_j + \eta)^2 + \frac{1}{2m^+} \sum_{i=1}^{m^+} (\epsilon_i + \eta)^2 \right\}. \quad (5.71)$$

The normalizing factor of 2 in both the false negative and false positive terms accounts for the differentiation factor resulted from the two square terms.

Total Error Rate Based Learning

Recall that $\epsilon_j := g(\mathbf{x}_j^-, \mathbf{w}) - \tau$, $j = 1, 2, \dots, m^-$ and $\epsilon_i := \tau - g(\mathbf{x}_i^+, \mathbf{w}) - \Delta$, $i = 1, 2, \dots, m^+$. Given a linear predictor $g(\mathbf{x}_i, \mathbf{w}) = \mathbf{w}^T \mathbf{p}(\mathbf{x}_i)$ and considering a weight decay regularization by adding $\frac{b}{2} \|\mathbf{w}\|_2^2$, we have

$$\min_{\mathbf{w}} \text{TER}(\mathbf{w}) \approx \min_{\mathbf{w}} \left\{ \frac{b}{2} \|\mathbf{w}\|_2^2 + \frac{1}{2m^-} \sum_{j=1}^{m^-} [(\mathbf{w}^T \mathbf{p}(\mathbf{x}_j^-) - \tau) + \eta]^2 + \frac{1}{2m^+} \sum_{i=1}^{m^+} [(\tau - \mathbf{w}^T \mathbf{p}(\mathbf{x}_i^+)) + \eta]^2 \right\}, \quad (5.72)$$

where the value of b controls the amount of regularization.

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Total Error Rate Based Learning

Abbreviating the column basis vectors $\mathbf{p}_j = \mathbf{p}(\mathbf{x}_j^-) \in \mathbb{R}^{D+1}$ and $\mathbf{p}_i = \mathbf{p}(\mathbf{x}_i^+) \in \mathbb{R}^{D+1}$, the [solution for \$\mathbf{w}\$](#) which minimizes (5.72) can be written as

$$\hat{\mathbf{w}} = \left[b\mathbf{I} + \frac{1}{m^-} \sum_{j=1}^{m^-} \mathbf{p}_j \mathbf{p}_j^T + \frac{1}{m^+} \sum_{i=1}^{m^+} \mathbf{p}_i \mathbf{p}_i^T \right]^{-1} \left[\frac{(\tau - \eta)}{m^-} \sum_{j=1}^{m^-} \mathbf{p}_j + \frac{(\tau + \eta)}{m^+} \sum_{i=1}^{m^+} \mathbf{p}_i \right] \quad (5.75)$$

where $\mathbf{I}$ is an identity matrix of $(D + 1) \times (D + 1)$ size (recall that $D + 1$ is the total number of transformed feature terms).

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$$\left[\begin{array}{c} -T_{(n+)} \end{array} \right] \qquad \qquad \qquad \left[\begin{array}{c} -T_{(n-)} \end{array} \right]$$

— *III* —

Total Error Rate Based Learning

Multi-category Learning: Consider the multi-class problem with C output categories. An indicator matrix can be formed for the above TER formulation similar to that in LSE formulation. Assume a common setting for threshold (τ) and bias (η) throughout all C outputs. Subsequently, we have for each k th-class ($k = 1, 2, \dots, C$), a positive output $\mathbf{y}_k^+ = (\tau + \eta)\mathbf{1}_+ \in \mathbb{R}^{m^+}$ which corresponds to a k th-class indicator, and a negative output $\mathbf{y}_k^- = (\tau - \eta)\mathbf{1}_- \in \mathbb{R}^{m^-}$ which corresponds to a non- k th-class indicator. By packing the solutions with corresponding $\mathbf{P}_{k+}$ and $\mathbf{P}_{k-}$ matrices, equation (5.76) can be extended to multi-category as

$$\hat{\mathbf{W}} = [\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_C], \quad (5.78)$$

where

$$\hat{\mathbf{w}}_k = \left(b\mathbf{I} + \frac{1}{m_k^-} \mathbf{P}_{k-}^T \mathbf{P}_{k-} + \frac{1}{m_k^+} \mathbf{P}_{k+}^T \mathbf{P}_{k+} \right)^{-1} \times \left(\frac{1}{m_k^-} \mathbf{P}_{k-}^T \mathbf{y}_k^- + \frac{1}{m_k^+} \mathbf{P}_{k+}^T \mathbf{y}_k^+ \right),$$

$$k = 1, \dots, C.$$

Total Error Rate Based Learning

Total Error Rate Based Learning in the Dual Space: By defining two class-specific diagonal weighting matrices

$$\mathbf{M}_- = \begin{bmatrix} \frac{1}{m^-} & & & & 0 \\ & \ddots & & & \\ & & \frac{1}{m^-} & & \\ & & & 0 & \\ 0 & & & & \ddots & \\ & & & & & 0 \end{bmatrix} \begin{matrix} \left. \vphantom{\begin{matrix} \frac{1}{m^-} \\ \ddots \\ \frac{1}{m^-} \end{matrix}} \right\} m^- \\ \left. \vphantom{\begin{matrix} 0 \\ \ddots \\ 0 \end{matrix}} \right\} m^+ \end{matrix} \quad (5.79)$$

$$\mathbf{M}_+ = \begin{bmatrix} 0 & & & & 0 \\ & \ddots & & & \\ & & 0 & & \\ & & & \frac{1}{m^+} & \\ 0 & & & & \ddots & \\ & & & & & \frac{1}{m^+} \end{bmatrix} \begin{matrix} \left. \vphantom{\begin{matrix} 0 \\ \ddots \\ 0 \end{matrix}} \right\} m^- \\ \left. \vphantom{\begin{matrix} \frac{1}{m^+} \\ \ddots \\ \frac{1}{m^+} \end{matrix}} \right\} m^+ \end{matrix}, \quad (5.80)$$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

Total Error Rate Based Learning

By putting $\mathbf{w} = \mathbf{P}^T \mathbf{M} \alpha$ and using the derivation steps of the [dual ridge regression](#), we have

$$\hat{\alpha} = (b\mathbf{I} + \mathbf{P}\mathbf{P}^T \mathbf{M})^{-1} \mathbf{y}. \quad (5.83)$$

This $\hat{\alpha}$ can be substituted back into $\mathbf{w} = \mathbf{P}^T \mathbf{M} \alpha$ to give

$$\hat{\mathbf{w}} = \mathbf{P}^T \mathbf{M} (b\mathbf{I} + \mathbf{P}\mathbf{P}^T \mathbf{M})^{-1} \mathbf{y}. \quad (5.84)$$

For multi-category case, the solution for parameter estimation can be packed as:

$$\hat{\mathbf{W}} = [\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_C] \quad (5.85)$$

where

$$\hat{\mathbf{w}}_k = \mathbf{P}_k^T \mathbf{M}_k (b\mathbf{I} + \mathbf{P}_k \mathbf{P}_k^T \mathbf{M}_k)^{-1} \mathbf{y}_k, \quad k = 1, 2, \dots, C. \quad (5.86)$$

Here, it is noted that for each $\hat{\mathbf{w}}_k$, $k = 1, \dots, C$ the sample order of matrices $\mathbf{P}_k$, $\mathbf{M}_k$ needs to be congruence to the sample order of $\mathbf{y}_k$ in terms of their distinct set of indicating target labels for each $k \in \{1, 2, \dots, C\}$.

Example 5.4

Consider the following training data points for two categories,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}: \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

which correspond to labels $y : \{1, 0, 1, 0, 0\}$ respectively. Using a 3rd-order

polynomial model and the TER method, our goal is to predict the class labels of

$$\mathbf{x}_{t1} = \begin{bmatrix} 2.5 \\ 4 \end{bmatrix} \text{ (red 'triangle' in Fig. 2) and } \mathbf{x}_{t2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ (black 'triangle' in Fig. 2)}$$

which have true labels $y_{t1} = 0, y_{t2} = 1$ respectively.

Fig. 2 shows the decision boundary at threshold 0 according to the one-hot implementation of the binary target. This threshold is different from that in Example 3 which has an explicit threshold setting at 0.5. The Python codes for TER learning which can operate in either the Primal or the Dual modes are shown below.

Example 5.4

```

from sklearn.preprocessing import PolynomialFeatures
X = np.array([[2,2],[1,2],[1,1],[3,1],[2,3]])
Y = np.array([[1,0],[0,1],[1,0],[0,1],[0,1]])
Xt = np.array([[2.5,4],[1,0]])
Yt = np.array([[0,1],[1,0]])
poly = PolynomialFeatures(3)
P = poly.fit_transform(X)
Pt = poly.fit_transform(Xt)
W_dual = TERtrain(P,Y,0)
Yt_predict, Yt_class_predict = TERtest(Pt, Yt, W_dual)
M1, I1 = np.max(Yt, axis=1), np.argmax(Yt, axis=1)
M2, I2 = np.max(Yt_class_predict, axis=1), np.argmax(Yt_class_predict,
axis=1)
TestAccuracy = (np.mean(I1 == I2))*100
print(Yt_predict)
print(Yt_class_predict)
print(TestAccuracy)
[[-0.67427395  1.51920227]
 [ 1.17037166 -0.18958447]]
[[0.  1.]
 [1.  0.]]
100.0

```

Example 5.4 Cont'd

```

def TERtrain(P,Y,b=None,Sp=None,Sm=None):
    # Training by TER method Primal/Dual
    if b is None and Sp is None and Sm is None:
        # 'b' and 'Sp, Sm' scaling not given
        Sp=1; Sm=1; b = 1e-4 # Use default settings
    elif Sp is None and Sm is None:
        # only 'Sp, Sm' scaling not given
        Sp=1; Sm=1
    m,K = Y.shape
    W = np.zeros((P.shape[1], K))
    for jj in range(K):
        max_value = np.max(Y[:, jj])
        min_value = np.min(Y[:, jj])
        Mp, D = P[Y[:,jj]==max_value,:].shape
        Mm, D = P[Y[:,jj]==min_value,:].shape
        Mvector = np.ones(m)
        Mvector[Y[:,jj]==min_value] = 1/(Sm*Mm)
        Mvector[Y[:,jj]==max_value] = 1/(Sp*Mp)
        M = np.diag(Mvector)
        if m>=D: # Primal
            II = np.eye(D,D)
            W[:,jj] = inv(P.T @ M @ P + b*II) @ P.T @ M @ Y[:,jj]
        else: # Dual
            II = np.eye(m,m)
            W[:,jj] = P.T @ inv(M @ P @ P.T + b*II) @ M @ Y[:,jj]
    return W

```

Example 5.4 Cont'd

```

def TERtest(Pt, Yt, W):
    #- prediction -%
    Yt_out_pred = Pt @ W
    nrow, ncol = Yt.shape
    if ncol == 1:
        #- Binary -%
        Yt_class_pred = np.zeros((nrow, 1))
        Yt_class_pred[Yt_out_pred >= (np.max(Yt) + np.min(Yt)) / 2] = np.max(Yt)
        Yt_class_pred[Yt_out_pred < (np.max(Yt) + np.min(Yt)) / 2] = np.min(Yt)
        TestAccuracy = 100 * len(np.where(Yt == Yt_class_pred)[0]) / nrow
    else:
        #- one-versus-all -%
        Yt_class_pred = np.zeros((nrow, ncol))
        for jj in range(nrow):
            jindx = np.argmax(Yt_out_pred[jj, :], np.argmax(Yt_out_pred[jj, :]))
            Yt_class_pred[jj, jindx] = 1
        M1, I1 = np.max(Yt, axis=1), np.argmax(Yt, axis=1)
        M2, I2 = np.max(Yt_class_pred, axis=1), np.argmax(Yt_class_pred, axis=1)
        TestAccuracy = (np.mean(I1 == I2)) * 100
    return Yt_out_pred, Yt_class_pred, TestAccuracy

```

Example 5.4 Cont'd

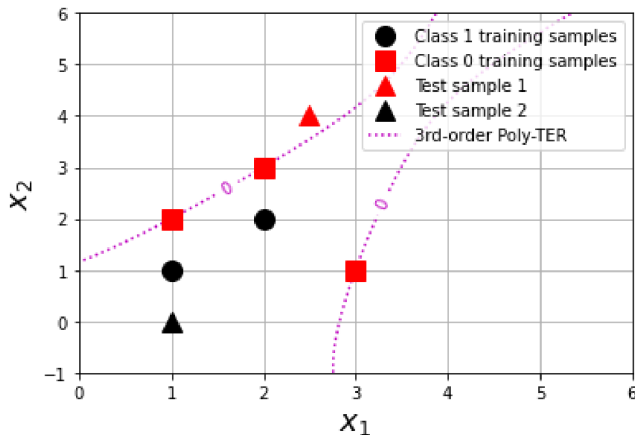


Figure 2: Learning decision contour at level [0].

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- Apart from the error rate based cost function measured at single operating threshold, the receiver operating characteristic (ROC) curve provides an overview of error costs over all operating thresholds which can also be utilized for learning.
- However, the ROC gives a range of operating points instead of a scalar value, it is not convenient for optimization.
- The Area Under the ROC Curve (AUC) is thus a natural alternative as the learning cost function for classifier learning concerning all operating points.

AUC Based Learning

For binary data samples indexed by $i = 1, 2, \dots, m^+$ and $j = 1, 2, \dots, m^-$ in respective categories, let us denote the difference between the predictor values of a positive sample $\mathbf{x}_i^+$ and that of a negative sample $\mathbf{x}_j^-$ by $\xi_{ij} = g(\mathbf{x}_i^+) - g(\mathbf{x}_j^-)$. Here, $\xi_{ij} > 0$ indicates that the positive sample $\mathbf{x}_i^+$ and the negative sample $\mathbf{x}_j^-$ are correctly ranked by the predictor g . Consider the Heaviside step function given by

$$u(\xi) = \begin{cases} 1, & \text{if } \xi > 0, \\ 0.5 & \text{if } \xi = 0, \\ 0, & \text{if } \xi < 0. \end{cases} \quad (5.87)$$

Then, the AUC for $m = m^+ + m^-$ training examples can be expressed as:

$$\begin{aligned} \text{AUC}(\mathbf{w}, \mathbf{x}) &= \frac{1}{m^+ m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} u(\xi_{ij}) \\ &= \frac{1}{m^+ m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} u(g(\mathbf{x}_i^+) - g(\mathbf{x}_j^-)), \end{aligned} \quad (5.88)$$

which is the fraction of positive-negative pairs that are ranked correctly by predictor $g(\cdot)$. The value in (5.88) is also known as the Wilcoxon-Mann-Whitney statistic.

AUC Based Learning

The **AUC can be maximized with respect to $\mathbf{w}$** of the parametric predictor $g(\mathbf{x}, \mathbf{w}) = \mathbf{w}^T \mathbf{p}(\mathbf{x})$ for effective classification considering the entire operating threshold range. Alternatively, the following equivalent minimization of the *Area Above the ROC Curve* (AAC) can be used:

$$\min_{\mathbf{w}} \text{AAC}(\mathbf{w}, \mathbf{x}) = \min_{\mathbf{w}} \frac{1}{m^+ m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} u(-\xi_{ij}) \quad . \quad (5.89)$$

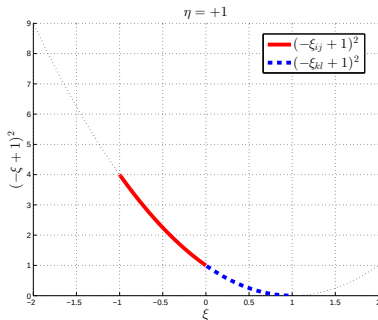
Similar to the TER approximation, when $g(\cdot)$ is in linear parametric form, the minimization problem given by (5.89) **can be solved by a quadratic approximation** to the non-differentiable step function $u(\cdot)$ with a weight decay such as

$$\begin{aligned} & \min_{\mathbf{w}} \text{AAC}(\mathbf{w}, \mathbf{x}) \\ &= \min_{\mathbf{w}} \left\{ \frac{b}{2} \|\mathbf{w}\|_2^2 + \frac{1}{2 m^+ m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \left[\left(\mathbf{p}(\mathbf{x}_j^-) - \mathbf{p}(\mathbf{x}_i^+) \right)^T \mathbf{w} + \eta \right]^2 \right\} . \end{aligned} \quad (5.90)$$

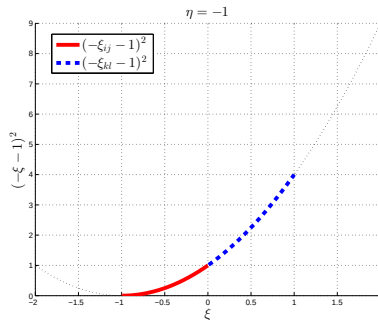
AUC Based Learning

- Fig. 3(a) shows an illustration of the quadratic output values of those wrongly ranked data ($\xi_{ij} < 0$, which is indicated by solid-line) being higher than those generated from correctly ranked data ($\xi_{kl} > 0$, which is indicated by dashed-line), i.e., $(-\xi_{ij} + \eta)^2 > (-\xi_{kl} + \eta)^2$ whenever $\xi_{ij} < 0$ and $\xi_{kl} > 0$ when $\eta = +1$.
- Conversely, when $\eta = -1$ in Fig. 3(b), those wrongly ranked data ($\xi_{ij} < 0$, indicated by solid-line) give lower quadratic output values $(-\xi_{ij} + \eta)^2$ than those given by $(-\xi_{kl} + \eta)^2$, which are generated by the correctly ranked data ($\xi_{kl} > 0$, indicated by dashed-line). This means that only one arm of the quadratic function has been utilized for monotonic ranking approximation.

AUC Based Learning



(a) $\eta = +1$: solid-line $>$ dashed-line



(b) $\eta = -1$: solid-line $<$ dashed-line

Figure 3: (a) $\eta = +1$: $(-\xi_{ij} + \eta)^2 > (-\xi_{kl} + \eta)^2$ whenever $\xi_{ij} < 0$ and $\xi_{kl} > 0$; (b) $\eta = -1$: $(-\xi_{ij} + \eta)^2 < (-\xi_{kl} + \eta)^2$ whenever $\xi_{ij} < 0$ and $\xi_{kl} > 0$.

AUC Based Learning

Under this setting, the minimization in (5.90) can be solved by setting its gradient with respect to $\mathbf{w}$ equals to zero, giving

$$\hat{\mathbf{w}} = \left[b\mathbf{I} + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\mathbf{p}_j^- - \mathbf{p}_i^+) (\mathbf{p}_j^- - \mathbf{p}_i^+)^T \right]^{-1} \times \left[\frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\mathbf{p}_j^- - \mathbf{p}_i^+) \right], \quad (5.91)$$

with the projection column vectors being abbreviated by $\mathbf{p}_j^- := \mathbf{p}(\mathbf{x}_j^-)$ and $\mathbf{p}_i^+ := \mathbf{p}(\mathbf{x}_i^+)$.

AUC Based Learning

The above solution for **AUC or AAC is optimal with respect to the full range of operating threshold decision points**. Hence, there is no explicit threshold being defined in (5.91). However, **in practice, a specific choice of threshold is often needed for decision making**. To do this, an **optimal threshold with respect to the total error rate** can be obtained by minimizing the approximated TER (5.71) with respect to τ and utilizing the above AAC solution (5.91), giving

$$\tau = \frac{1}{2m^-} \sum_{j=1}^{m^-} \hat{\mathbf{w}}^T \mathbf{p}_j^- + \frac{1}{2m^+} \sum_{i=1}^{m^+} \hat{\mathbf{w}}^T \mathbf{p}_i^+. \quad (5.92)$$

In summary, the AAC minimizer (5.91) provides an optimal solution for all operating classification thresholds. This is useful when the threshold needs to be adjusted without compromising the optimality in terms of the overall operating performance. To cater for specific operating requirements, an optimal threshold with respect to minimizing the TER can nevertheless be chosen based on (5.92).

Example 5.5

Consider the following training data points for two categories,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

which correspond to labels $y : \{1, 0, 1, 0, 0\}$ respectively. Our goal here is use AUC learning to

predict the class labels of $\mathbf{x}_{t1} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ (red ‘triangle’ in Fig. 2) and $\mathbf{x}_{t2} = \begin{bmatrix} 3 \\ 0.3 \end{bmatrix}$ (black

‘triangle’ in Fig. 2) which have true labels $y_{t1} = 0, y_{t2} = 1$ respectively.

- Based on the structural resemblance among the LSE, TER and AUC solutions, it is observed that these classifiers can be linked together by simple data manipulation.
- Here, we introduce an additive form of linearly transformed data at feature level for the AUC formulation.
- By mere variation of coefficient settings of the additive terms, we show that several linear classifiers can be interrelated.
- The knowledge of this relationship leads to a generalized classifier which allows variation of its coefficient settings beyond the above known settings.

Data Transformation

Consider each sample of input feature vector given by $\mathbf{x} \in \mathbb{R}^d$ with a corresponding target label given by $y \in \{0, 1\}$, a **transformation on the data** $\mathbf{x}$ can be performed as follows:

$$\tilde{\mathbf{x}} = \mathbf{T}\mathbf{x} + \mathbf{a}, \quad (5.93)$$

where $\mathbf{a} = [a_1, a_2, \dots, a_d]^T \in \mathbb{R}^d$ is a column vector for translation and $\mathbf{T}$ is a $d \times d$ matrix.

Next consider m number of training samples given by $(\mathbf{x}_i, y_i)$, where $i = 1, 2, \dots, m$. Suppose there are K **distinct scaling matrices** $\mathbf{T}_k$ and **translating vectors** $\mathbf{a}_k$, $k = 1, 2, \dots, K$. This gives rise to $m \times K$ training data samples:

$$\tilde{\mathbf{x}}_{ik} = \mathbf{T}_k \mathbf{x}_i + \mathbf{a}_k, i = 1, \dots, m, k = 1, \dots, K. \quad (5.94)$$

In a similar fashion, we could also work at the **transformed projection vectors** of the generalized linear models:

$$\tilde{\mathbf{p}}(\mathbf{x}) = \mathbf{T}\mathbf{p}(\mathbf{x}) + \mathbf{a}, \quad (5.95)$$

where $\mathbf{p}(\mathbf{x}) \in \mathbb{R}^{D+1}$ projects $\mathbf{x}$ into a nonlinear hyperplane, $\mathbf{T}$ is $(D+1) \times (D+1)$ matrix and $\mathbf{a} \in \mathbb{R}^{D+1}$.

d

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$$\tilde{\mathbf{p}}_{ki}^+ = \tilde{\mathbf{p}}_k(\mathbf{x}_i^+) = \mathbf{T}_k \mathbf{p}(\mathbf{x}_i^+) + \mathbf{a}_k,$$

1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 26

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$$\min_{\mathbf{w}} \text{AAC}(\mathbf{w}, \mathbf{x}^+, \mathbf{x}^-) = \min_{\mathbf{w}} \left\{ \frac{b}{2} \|\mathbf{w}\|_2^2 + \frac{1}{2m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \sum_{k=1}^K \left[(\tilde{\mathbf{p}}_{kj}^- - \tilde{\mathbf{p}}_{ki}^+)^T \mathbf{w} + \eta \right]^2 \right\}. \quad (5.98)$$

Data Transformation

Then, according to the AUC solution (26), the solution $\tilde{\mathbf{w}}_{\text{TAUC}}$ which minimizes the **Transformed AUC (TAUC)** in (5.98) is given by

$$\tilde{\mathbf{w}}_{\text{TAUC}} = \left[b\mathbf{I} + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \sum_{k=1}^K (\tilde{\mathbf{p}}_{kj}^- - \tilde{\mathbf{p}}_{ki}^+) (\tilde{\mathbf{p}}_{kj}^- - \tilde{\mathbf{p}}_{ki}^+)^T \right]^{-1} \\ \times \left[\frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \sum_{k=1}^K (\tilde{\mathbf{p}}_{kj}^- - \tilde{\mathbf{p}}_{ki}^+) \right]. \quad (5.99)$$

Based on this generalized solution, we shall observe the relationships among several existing binary classifiers and predictors in the following subsections.

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Between TAUC and AUC Solutions

When $K = 2$ (i.e., with two scaling transformations) and $\mathbf{a}_k = \mathbf{0}$ (which means no translation), the AUC solution that incorporates only scaling transformation can be written as:

$$\begin{aligned} \tilde{\mathbf{w}} = & \left[b\mathbf{I} + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\beta_1 \mathbf{p}_j^- - \gamma_1 \mathbf{p}_i^+) (\beta_1 \mathbf{p}_j^- - \gamma_1 \mathbf{p}_i^+)^T \right. \\ & \left. + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\beta_2 \mathbf{p}_j^- - \gamma_2 \mathbf{p}_i^+) (\beta_2 \mathbf{p}_j^- - \gamma_2 \mathbf{p}_i^+)^T \right]^{-1} \\ & \times \left[\frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\beta_1 \mathbf{p}_j^- - \gamma_1 \mathbf{p}_i^+) + \frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\beta_2 \mathbf{p}_j^- - \gamma_2 \mathbf{p}_i^+) \right]. \quad (5.101) \end{aligned}$$

Here we see that the AUC solution (5.91) becomes a special case of (5.101) when $\beta_1 = \beta_2 = \gamma_1 = \gamma_2 = 1$. In other words, the TAUC solution generalizes the AUC solution by considering scaled data in the learning.

Between TAUC and TER Solutions

Now, consider another specific case for $K = 2$ with $\gamma_1 = \beta_2 = 0$ and $\beta_1 = \gamma_2 = 1$, we have

$$\begin{aligned}\tilde{\mathbf{w}} &= \left[b\mathbf{I} + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\beta_1 \mathbf{p}_j^-) (\beta_1 \mathbf{p}_j^-)^T + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (-\gamma_2 \mathbf{p}_i^+) (-\gamma_2 \mathbf{p}_i^+)^T \right]^{-1} \\ &\quad \times \left[\frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (\beta_1 \mathbf{p}_j^-) + \frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} (-\gamma_2 \mathbf{p}_i^+) \right] \\ &= \left[b\mathbf{I} + \frac{1}{m^-} \sum_{j=1}^{m^-} \mathbf{p}_j^- \mathbf{p}_j^{-T} + \frac{1}{m^+} \sum_{i=1}^{m^+} \mathbf{p}_i^+ \mathbf{p}_i^{+T} \right]^{-1} \left[\frac{0-\eta}{m^-} \sum_{j=1}^{m^-} \mathbf{p}_j^- + \frac{0+\eta}{m^+} \sum_{i=1}^{m^+} \mathbf{p}_i^+ \right].\end{aligned}\tag{5.102}$$

Except for a default zero threshold ($\tau = 0$), the above solution (5.102) is similar to that of TER.

Between TAUC and FLD Solutions

Consider, the [scaled version of \(5.101\)](#) plus a global translation for each class, i.e.,

$$\tilde{p}_k(\mathbf{x}) = ((1 - y)\beta_k + y\gamma_k)\mathbf{I}_{D+1}\mathbf{p}(\mathbf{x}) + (1 - y)\mathbf{u}_k + y\mathbf{v}_k, \text{ where } k = 1, 2,$$

for each sample pair $(\mathbf{x}, y)$, we get:

$$\begin{aligned} \tilde{\mathbf{w}} = & \left[b\mathbf{I} + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \left((\beta_1\mathbf{p}_j^- - \mathbf{u}_1) - (\gamma_1\mathbf{p}_i^+ - \mathbf{v}_1) \right) \left((\beta_1\mathbf{p}_j^- - \mathbf{u}_1) - (\gamma_1\mathbf{p}_i^+ - \mathbf{v}_1) \right)^T \right. \\ & \left. + \frac{1}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \left((\beta_2\mathbf{p}_j^- - \mathbf{u}_2) - (\gamma_2\mathbf{p}_i^+ - \mathbf{v}_2) \right) \left((\beta_2\mathbf{p}_j^- - \mathbf{u}_2) - (\gamma_2\mathbf{p}_i^+ - \mathbf{v}_2) \right)^T \right]^{-1} \\ & \times \left[\frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \left((\beta_1\mathbf{p}_j^- - \mathbf{u}_1) - (\gamma_1\mathbf{p}_i^+ - \mathbf{v}_1) \right) + \frac{-\eta}{m^+m^-} \sum_{i=1}^{m^+} \sum_{j=1}^{m^-} \left((\beta_2\mathbf{p}_j^- - \mathbf{u}_2) - (\gamma_2\mathbf{p}_i^+ - \mathbf{v}_2) \right) \right]. \end{aligned} \quad (5.103)$$

Between TAUC and FLD Solutions

When $b = \gamma_1 = \beta_2 = 0$, $\beta_1 = \gamma_2 = 1$ and $\eta = 1/2$, by setting $\mathbf{u}_2 = \mathbf{v}_1 = \mathbf{0}$, $\mathbf{u}_1 = \frac{1}{m^-} \sum_{j=1}^{m^-} \mathbf{p}_i^- = \boldsymbol{\mu}_-$ (mean of vectors $\mathbf{p}_i^-$) while $\mathbf{v}_2 = \frac{1}{m^+} \sum_{j=1}^{m^+} \mathbf{p}_j^+ = \boldsymbol{\mu}_+$ (mean of vectors $\mathbf{p}_j^+$), we arrive at

$$\tilde{\mathbf{w}} = \left[\frac{1}{m^-} \sum_{j=1}^{m^-} (\mathbf{p}_j^- - \boldsymbol{\mu}_-)(\mathbf{p}_j^- - \boldsymbol{\mu}_-)^T + \frac{1}{m^+} \sum_{i=1}^{m^+} (\mathbf{p}_i^+ - \boldsymbol{\mu}_+)(\mathbf{p}_i^+ - \boldsymbol{\mu}_+)^T \right]^{-1} \times (\boldsymbol{\mu}_+ - \boldsymbol{\mu}_-). \quad (5.104)$$

In particular, when $m^- = m^+ = 1$, the above solution is analogous to Fisher Linear Discriminant analysis (FLD, see [Duda et al., 2001], p.120), but with embedding of nonlinear features within basis expansion vector $\mathbf{p}$ (i.e., $\mathbf{p}(\mathbf{x})$ instead of $\mathbf{x}$). Hence the solution to the transformed AUC space can be generalized to contain the FLD solution. Moreover, the FLD solution can be considered a centered version of the TER solution.

Between TER and LSE Solutions

Utilizing the [weight matrices](#) $\mathbf{M}_-$, $\mathbf{M}_+$ and without considering the weight decay regularization, the TER solution (5.102) can be written using matrix notation, i.e.

$$\begin{aligned}\tilde{\mathbf{w}} &= [\mathbf{P}^T (\mathbf{M}_- + \mathbf{M}_+) \mathbf{P}]^{-1} [\mathbf{P}^T (\mathbf{M}_- + \mathbf{M}_+) \mathbf{y}] \\ &= (\mathbf{P}^T \mathbf{M} \mathbf{P})^{-1} \mathbf{P}^T \mathbf{M} \mathbf{y},\end{aligned}\tag{5.105}$$

where $\mathbf{M} = \mathbf{M}_- + \mathbf{M}_+$ and elements of $\mathbf{P}$ being ordered according to the two classes. The column vector $\mathbf{y}$ is the target vector given by

$$\left[\frac{(0 - \eta)}{m^-}, \dots, \frac{(0 - \eta)}{m^-}, \frac{(0 + \eta)}{m^+}, \dots, \frac{(0 + \eta)}{m^+} \right]^T.\tag{5.106}$$

Hence, the TER solution can be further related to a [Weighted Least Squares](#) (Weighted-LS) solution. Together with the well-known relationship between LS and Weighted-LS, this observation shows that the AUC in the scaling space (Equation (5.101)) provides a generalized framework for estimation based on [TER](#), [Weighted-LS](#) and [LS](#).

Relationship with Bayesian Inference

In statistics, the regression model is often analyzed within the context of Bayesian inference with particular assumption of likelihood and prior probability distributions. Consider a data set of input-target pairs $(\mathbf{x}_i, y_i)$, $i = 1, 2, \dots, m$ and assume that the targets are samples from the **model with additive noise e_i** :

$$y_i = g(\mathbf{x}_i, \mathbf{w}) + e_i = \mathbf{x}_i^T \mathbf{w} + e_i, \quad (5.107)$$

where $e_i = y_i - g(\mathbf{x}_i, \mathbf{w})$ being independent samples drawn from a certain noise process which is assumed to be **Gaussian with zero mean and variance σ^2** denoted as $\mathcal{N}(0, \sigma^2)$. Assume independence of samples y_i , the corresponding likelihood of the complete data set can be written as:

$$\begin{aligned} p(\mathbf{y}|\mathbf{X}, \mathbf{w}) &= \prod_{i=1}^m p(y_i|\mathbf{x}_i, \mathbf{w}) \\ &= (2\pi\sigma^2)^{-m/2} \exp \left\{ -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w}) \right\}, \quad (5.108) \end{aligned}$$

where $\mathbf{y}$, $\mathbf{X}$ and $\mathbf{w}$ are defined similarly to those stacked symbols in LSE.

Relationship with Bayesian Inference

This likelihood function for the data is next combined with a prior belief (frequently chosen again, to be Gaussian such as $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Sigma_p)$) about the parameters according to Bayes' theorem to yield the posterior belief about the parameter $\mathbf{w}$:

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \sim \mathcal{N}(\bar{\mathbf{w}}, \Sigma), \quad (5.109)$$

where $\bar{\mathbf{w}} = \frac{1}{\sigma^2} (\frac{1}{\sigma^2} \mathbf{P}^T \mathbf{P} + \Sigma_p^{-1})^{-1} \mathbf{P}^T \mathbf{y}$ and $\Sigma = (\frac{1}{\sigma^2} \mathbf{P}^T \mathbf{P} + \Sigma_p^{-1})^{-1}$.

Relationship with Bayesian Inference

Recall the TER formulation with quadratic approximation, which can be written in residual form as

$$\begin{aligned} \text{TER} &= \frac{1}{m^-} \sum_{j=1}^{m^-} L(\epsilon_j) + \frac{1}{m^+} \sum_{i=1}^{m^+} L(\epsilon_i) \\ &\approx \frac{1}{m^-} \sum_{j=1}^{m^-} (\epsilon_j + \eta)^2 + \frac{1}{m^+} \sum_{i=1}^{m^+} (\epsilon_i + \eta)^2, \end{aligned} \quad (5.110)$$

where $\epsilon_j = g(\mathbf{x}_j^-) - \tau$ and $\epsilon_i = \tau - g(\mathbf{x}_i^+)$ can both be assumed Gaussian with independency among samples within and between the two classes. Consider a linear projection model for g and let $y_j^- = \tau - \eta$ and $y_i^+ = \tau + \eta$, the above equation can be rewritten as:

$$\text{TER} \approx \frac{1}{m^-} \sum_{j=1}^{m^-} \left((\mathbf{x}_j^-)^T \mathbf{w} - y_j^- \right)^2 + \frac{1}{m^+} \sum_{i=1}^{m^+} \left(y_i^+ - (\mathbf{x}_i^+)^T \mathbf{w} \right)^2. \quad (5.111)$$

Country	Year	Value
China	2000	1.00
China	2001	1.00
China	2002	1.00
China	2003	1.00
China	2004	1.00
China	2005	1.00
China	2006	1.00
China	2007	1.00
China	2008	1.00
China	2009	1.00
China	2010	1.00
China	2011	1.00
China	2012	1.00
China	2013	1.00
China	2014	1.00
China	2015	1.00
China	2016	1.00
China	2017	1.00
China	2018	1.00
China	2019	1.00
China	2020	1.00
China	2021	1.00
China	2022	1.00
China	2023	1.00
China	2024	1.00
China	2025	1.00
China	2026	1.00
China	2027	1.00
China	2028	1.00
China	2029	1.00
China	2030	1.00
China	2031	1.00
China	2032	1.00
China	2033	1.00
China	2034	1.00
China	2035	1.00
China	2036	1.00
China	2037	1.00
China	2038	1.00
China	2039	1.00
China	2040	1.00
China	2041	1.00
China	2042	1.00
China	2043	1.00
China	2044	1.00
China	2045	1.00
China	2046	1.00
China	2047	1.00
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China	2071	1.00
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China	2074	1.00
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China	2091	1.00
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China	2093	1.00
China	2094	1.00
China	2095	1.00
China	2096	1.00
China	2097	1.00
China	2098	1.00
China	2099	1.00
China	2100	1.00
China	2101	1.00
China	2102	1.00
China	2103	1.00
China	2104	1.00
China	2105	1.00
China	2106	1.00
China	2107	1.00
China	2108	1.00
China	2109	1.00
China	2110	1.00
China	2111	1.00
China	2112	1.00
China	2113	1.00
China	2114	1.00
China	2115	1.00
China	2116	1.00
China	2117	1.00
China	2118	1.00
China		

Optimization of TER by quadratic approximation in (5.111) can thus be treated as a *dual-Gaussian regression*:

$$\begin{aligned}
p(\mathbf{y}|\mathbf{X}, \mathbf{w}) &= \left(\prod_{j=1}^{m^-} p(y_j^- | \mathbf{x}_j^-, \mathbf{w}) \right) \left(\prod_{i=1}^{m^+} p(y_i^+ | \mathbf{x}_i^+, \mathbf{w}) \right) \\
&= (2\pi\sigma_-^2)^{-m^-/2} \exp \left\{ -\frac{1}{2\sigma_-^2} (\mathbf{y}^- - \mathbf{X}_- \mathbf{w})^T (\mathbf{y}^- - \mathbf{X}_- \mathbf{w}) \right\} \\
&\quad \times (2\pi\sigma_+^2)^{-m^+/2} \exp \left\{ -\frac{1}{2\sigma_+^2} (\mathbf{y}^+ - \mathbf{X}_+ \mathbf{w})^T (\mathbf{y}^+ - \mathbf{X}_+ \mathbf{w}) \right\}.
\end{aligned} \tag{5.112}$$

Relationship with Bayesian Inference

By Bayes' rule, the posterior probability of parameter estimate can be written as

$$\begin{aligned}
 p(\mathbf{w}|\mathbf{y}, \mathbf{X}) &= \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y}|\mathbf{X})} \\
 &\propto \exp \left\{ -\frac{1}{2\sigma_-^2} (\mathbf{y}^- - \mathbf{X}_- \mathbf{w})^T (\mathbf{y}^- - \mathbf{X}_- \mathbf{w}) \right\} \\
 &\quad \times \exp \left\{ -\frac{1}{2\sigma_+^2} (\mathbf{y}^+ - \mathbf{X}_+ \mathbf{w})^T (\mathbf{y}^+ - \mathbf{X}_+ \mathbf{w}) \right\} \exp \left(-\frac{1}{2} \mathbf{w}^T \Sigma_p^{-1} \mathbf{w} \right).
 \end{aligned}
 \tag{5.113}$$

[illegible]

$$p(\mathbf{w}|\mathbf{y}, \mathbf{X}) \propto \exp\left(-\frac{1}{2}(\mathbf{w} - \bar{\mathbf{w}})^T \left(\frac{1}{\sigma_-^2} \mathbf{X}_-^T \mathbf{X}_- + \frac{1}{\sigma_+^2} \mathbf{X}_+^T \mathbf{X}_+ + \Sigma_P^{-1}\right) (\mathbf{w} - \bar{\mathbf{w}})\right), \quad (5.114)$$

[illegible]

$$\bar{\mathbf{w}} = \left(\frac{1}{\sigma_-^2} \mathbf{X}_-^T \mathbf{X}_- + \frac{1}{\sigma_+^2} \mathbf{X}_+^T \mathbf{X}_+ + \Sigma_p^{-1} \right)^{-1} \left(\frac{1}{\sigma_-^2} \mathbf{X}_-^T \mathbf{y}_- + \frac{1}{\sigma_+^2} \mathbf{X}_+^T \mathbf{y}_+ \right), \quad (5.115)$$

.....

$$\Sigma = (\frac{1}{\sigma_-^2} \mathbf{X}_-^T \mathbf{X}_- + \frac{1}{\sigma_+^2} \mathbf{X}_+^T \mathbf{X}_+ + \Sigma_p^{-1})^{-1}. \quad (5.116)$$

1. *Journal of the American Medical Association*, 2000; 283: 2689-2695.

Summary of Classifier Relationships

Table 1: Summary of classifiers and settings

		class-size, reg, offset				scaling				translation			
	Classifier	m^+	m^-	b	η	β_1	β_2	γ_1	γ_2	$\mathbf{u}_1$	$\mathbf{u}_2$	$\mathbf{v}_1$	$\mathbf{v}_2$
1.	FLD	1	1	b	1/2	1	0	0	1	$\boldsymbol{\mu}_+$	$\mathbf{0}$	$\mathbf{0}$	$\boldsymbol{\mu}_-$
2.	LSE	1	1	b	1/2	1	0	0	1	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$
3.	TER	m^+	m^-	b	1/2	1	0	0	1	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$
4.	AUC	m^+	m^-	b	1/2	1	1	1	1	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$
5.	Novel	m^+	m^-	b	1/2	r	r	r	r	r	r	r	r

m^+, m^- : sizes of positive class and negative class training samples respectively,

b : weight-decay regularization parameter which is chosen as 10^{-4} ,

η, r : η is quadratic offset as seen in (5.103), r is random number,

$\boldsymbol{\mu}_+, \boldsymbol{\mu}_-$: $\boldsymbol{\mu}_+ = \frac{1}{m^+} \sum_{j=1}^{m^+} \mathbf{p}_i^+$ which is mean of positive class features,
 $\boldsymbol{\mu}_- = \frac{1}{m^-} \sum_{j=1}^{m^-} \mathbf{p}_j^-$ which is mean of negative class features,

$\beta_1, \beta_2, \gamma_1, \gamma_2$: scaling parameters (scalars),

$\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_1, \mathbf{v}_2$: translational parameters (vectors).

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Thank you!



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